

Clinical Investigation: Central Nervous System Cancer

Risk Factors for Neovascular Glaucoma After Proton Beam Therapy of Uveal Melanoma: A Detailed Analysis of Tumor and Dose—Volume Parameters

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Summary

Neovascular glaucoma (NVG) is a serious complication after particle therapy for uveal melanoma. Review of our prospectively maintained database showed that the overall NVG rate was lower than observed historically. Predictors of NVG included tumor height, diameter, and anterior and posterior critical structure doses. Clinically these tumor and dose—volume parameters may help in evaluating NVG risk, though individual variability exists, and when appropriate, critical structure—sparing techniques may be used to reduce NVG incidence.

Purpose: To determine neovascular glaucoma (NVG) incidence and identify contributing tumor and dosing factors in uveal melanoma patients treated with proton beam radiation therapy (PBRT).

Methods and Materials: A total of 704 PBRT patients treated by a single surgeon (DHC) for uveal melanoma (1996-2010) were reviewed for NVG in our prospectively maintained database. All patients received 56 GyE in 4 fractions. Median follow-up was 58.3 months. Analyses included the Kaplan-Meier method to estimate NVG distributions, univariate log—rank tests, and Cox's proportional hazards multivariate analysis using likelihood ratio tests to identify independent risk factors of NVG among patient, tumor, and dose—volume histogram parameters.

Results: The 5-year PBRT NVG rate was 12.7% (95% confidence interval [CI] 10.2%-15.9%). The 5-year rate of enucleation due to NVG was 4.9% (95% CI 3.4%-7.2%). Univariately, the NVG rate increased significantly with larger tumor diameter ($P < .0001$), greater height ($P < .0001$), higher T stage ($P < .0001$), and closer proximity to the disc ($P = .002$). Dose—volume histogram analysis revealed that if $>30\%$ of the lens or ciliary body received $\geq 50\%$ dose (≥ 28 GyE), there was a higher probability of NVG ($P < .0001$ for both). Furthermore, if 100% of the disc or macula received ≥ 28 GyE, the NVG rate was higher ($P < .0001$ and $P = .03$, respectively). If both anterior and posterior doses were above specified cut points, NVG risk was highest ($P < .0001$). Multivariate analysis confirmed significant independent risk factors to include tumor height ($P < .0001$), age ($P < .0001$), %disc treated to $\geq 50\%$ Dose ($<100\%$ vs 100%) ($P = .0007$), larger tumor diameter ($P = .01$), %lens treated to $\geq 90\%$ Dose (0 vs $>0\%$ - 30% vs $>30\%$) ($P = .01$), and optic nerve length treated to $\geq 90\%$ Dose (≤ 1 mm vs >1 mm) ($P = .02$).

Conclusions: Our current PBRT patients experience a low rate of NVG and resultant enucleation compared with historical data. The present analysis shows that tumor height, diameter, and anterior as well as posterior critical structure dose—volume parameters may be used to predict NVG risk. © 2013 Elsevier Inc.

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Introduction

Uveal melanoma (UM), although a rare disease, is a malignancy that threatens both sight and life. Radiation therapy (RT) with plaques, stereotactic techniques, and charged particles, including proton (PBRT), helium, and carbon ion therapy, have been a mainstay of treatment (1-3). The physical characteristics of particles allow for uniform dose distribution, minimal scatter, and sharp dose falloff. Although overall long-term tumor control has been high with particles, treatment complications have been historically significant, particularly in terms of anterior segment side effects, such as cataract and neovascular glaucoma (NVG) (4-7). In a previous study of our helium ion patients, the 5-year NVG incidence was 35%, and most post-radiation enucleations were NVG related (8). At that time we postulated that reducing the radiation dose to the anterior chamber might result in a reduced incidence of NVG after charged particle therapy.

On the basis of these observations, the subsequently treated PBRT patients were treated using a beam delivery system designed to maximally spare the anterior segment by adjusting treatment-planning parameters, mainly gaze angle, and selectively limiting margins at critical structures as described below. In the present study we further investigate the factors that contribute to NVG risk, given our current beam delivery system. In particular, we analyze patient, tumor, and treatment parameters and dose-volume relationships for anterior and posterior critical structures to determine predictors of NVG incidence after PBRT.

Methods and Materials

Patient population

From January 1996 to December 2010, 766 UM patients from a single surgeon (DHC) were treated with PBRT at our institution with our current standard protocol dose of 56 gray equivalent (GyE; proton relative biological effectiveness = 1.1) in 4 fractions. Forty-seven patients were excluded owing to prior surgical resection and 15 patients because of previous ^{125}I plaque treatment, resulting in 704 evaluable patients. Patients were followed in a standardized manner including clinical evaluation, fundus photographs, and ultrasonography obtained approximately 3 months, 6 months, and then yearly after PBRT. A diagnosis of NVG was based on the presence of both increased intraocular pressure >21 mm and neovascularity at the angle. Patients diagnosed with NVG were evaluated for medical, anti-vascular endothelial growth factor, and/or photocoagulation treatment, and enucleation if symptoms and intraocular pressure could not be controlled otherwise. Ethics approval was obtained from the institutional research ethics board. Median follow-up was 58.3 months.

Radiation technique

Our PBRT delivery has been previously described (9, 10). Briefly, tantalum markers were sutured at 4 scleral points to outline the tumor base, and their spatial coordinates were obtained from 2 orthogonal x-ray films. All patients were immobilized with a customized face-mask and dental bite block. EYEPLAN software was used for treatment planning,

including eye and tumor modeling, gaze angle optimization, aperture design, beam parameter delineation, dose-volume histogram (DVH) analysis, marker alignment, and daily treatment positioning (11).

For each patient treatment, a 2.5-mm lateral aperture margin is placed around the tumor projection in the beam's-eye view, to account for possible microscopic tumor extension, minor patient setup error, eye movement during treatment, and beam penumbra. The aperture margin represents the 50% isodose line. Distal margin on the tumor range is 3 mm. Attempts to reduce the dose to the anterior critical structures were made as much as deemed safely possible and clinically appropriate. This was accomplished primarily by adjusting the gaze angle to minimize dose to the lens and/or ciliary body (CB). In addition, the gaze angle options were further analyzed to reduce the dose to the optic disc and/or macula if possible, with the goal of preserving vision. The aperture margin at the lens, macula, or disc in certain cases was focally reduced to 2 mm to reduce critical structure dose. In very select cases the distal margin was reduced to 2-2.5 mm to reduce the dose to the disc, macula, or nerve. Wedge filters were used in a limited fashion to significantly reduce the disc dose when appropriate.

Significant attention was given to eye position throughout the treatment via a closed circuit television camera system, and adjustments and/or repositioning were done if clinically necessary to ensure proper alignment. Custom retractors were used to remove the eyelids from the beam path as much as possible. If a significant amount of eyelid remained in the field, then the distal range was adjusted appropriately to compensate for the additional tissue. All patients were treated in a seated position with a single anterior port using a 67.5-MeV horizontal proton beam generated by a 76-inch isochronous cyclotron. Because of the treatment delivery setup, additional beam angles were not used to achieve anterior structure sparing.

Statistical analysis of NVG risk

The probability of NVG was estimated using the Kaplan-Meier method. Time to NVG was calculated from the beginning of RT to NVG occurrence or to the last ophthalmologic follow-up, enucleation, or death (whichever occurred first for those not developing NVG). Univariate variables included gender, age, diabetes, right/left eye, tumor location, height, largest tumor diameter (LTD), TNM size category (American Joint Committee on Cancer, 7th edition, 2010), distances to disc and fovea, CB involvement (CBI), and dose-volume relationships for the lens, CB, macula, disc, and optic nerve. Subsets were compared using the log-rank test.

Factors found to be predictive of NVG with a probability value <.10 with univariate methods were considered in a multivariate analysis to identify independent predictors using Cox's proportional hazards regression model. A forward stepwise approach was applied with the probability to include a variable in the model set at .05 and to remove a variable at .10. Significance was determined by the likelihood ratio test. For all analyses a probability value of <.05 was defined as significant. The analysis was conducted using Stata v8 (Stata Corp, College Station, TX).

Descriptive statistics for baseline features plus analysis of variance (ANOVA) methods to compare means and Pearson's correlation coefficients to evaluate linear relationships for continuous variables and χ^2 tests to assess independence between categorical variables were done.

Results

Baseline characteristics

Patient and tumor features are shown in Table 1. Of note, using the TNM size category we found no difference among the 4 subsets in mean age at the time of treatment (ANOVA, $P = .23$). The distance to the optic disc and macula were correlated with both the tumor height and LTD for this study cohort ($P < .0001$ for all correlations); specifically, the shorter the distance from the disc/macula, the smaller the height and diameter.

NVG rate

The cumulative incidence of postradiation NVG for proton-irradiated patients is shown in Figure 1a. Of the 704-patient cohort, 77 developed NVG, and 29 were enucleated because of NVG. For the 77 patients diagnosed with NVG, diagnosis occurred between 3.6 and 87 months after treatment, with a median of 22 months. The 5-year NVG rate in our PBRT cohort was 12.7% (95% confidence interval [CI] 10.2%-15.9%). The 5-year rate of enucleation due to NVG was 4.9% (95% CI 3.4%-7.2%; Fig. 1b).

Patient/tumor factors predictive of NVG

Table 2 and Figure 2 summarize the Kaplan-Meier estimated incidence of NVG by various patient and tumor factors. Significant predictors for increased NVG incidence included greater tumor height ($P < .0001$), larger tumor diameter ($P < .0001$), higher TNM size category ($P < .0001$), shorter distance to the disc ($P = .002$), and macular involvement ($P = .045$).

DVH factors predictive of NVG

The DVH data showed that both anterior as well as posterior critical structure dose-volume factors were significant predictors for NVG incidence. Specifically, when $>30\%$ of an anterior critical structure (lens or CB volume) received $\geq 50\%$ Dose (28

Table 1 Characteristics of patients/tumors treated with proton beam therapy	
Total patients	704
Gender (%) (male/female)	49.6/50.4
Treatment site (%) (right/left)	52.8/47.2
Age (y), median (range)	60 (13-94)
Tumor height (mm), mean $\pm$ SD (range)	5.2 $\pm$ 2.7 (1.3-18.3)
Largest tumor diameter (mm), mean $\pm$ SD [range]	10.7 $\pm$ 3.4 (2.0-23.7)
Tumor distance to fovea (mm), mean $\pm$ SD	3.9 $\pm$ 4.3
Tumor distance to optic disc (mm), mean $\pm$ SD	4.5 $\pm$ 4.2
Ciliary body involvement (%)	10.2
Diabetes at the time of diagnosis (%)	6.8
Follow-up (mo), median (range)	58.3 (6-194)

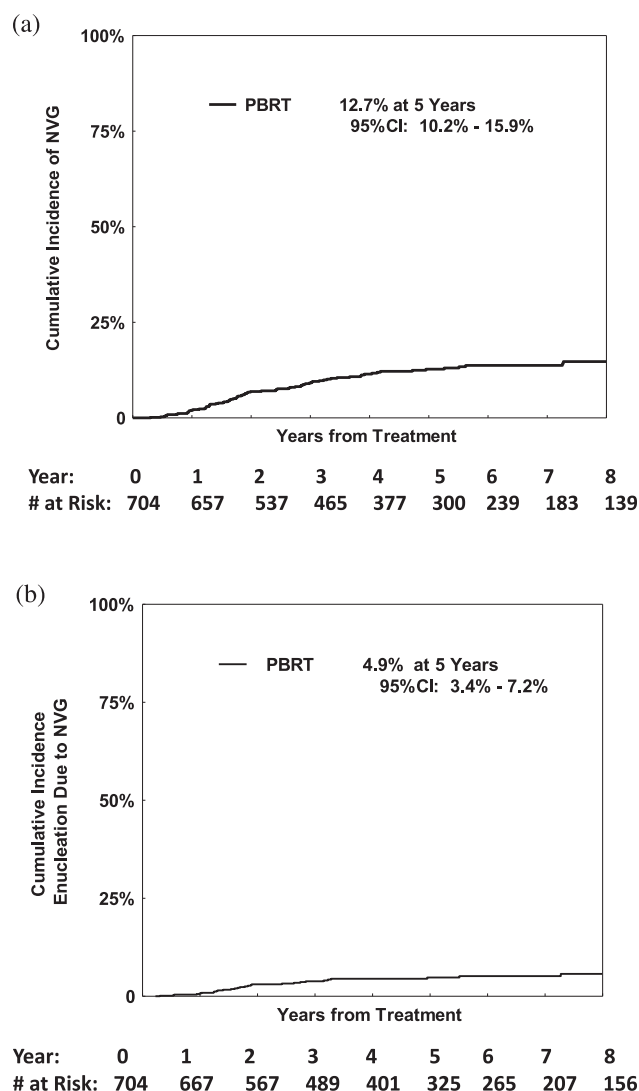


Fig. 1. Kaplan-Meier estimates after proton beam therapy (PBRT) of the (a) cumulative incidence of neovascular glaucoma (NVG); and (b) cumulative incidence of enucleation due to NVG.

GyE or higher), there was a substantially higher probability of NVG ($P < .0001$ for each). In addition, when 100% of a posterior structure (disc, nerve, or macula) received $\geq 50\%$ Dose, the NVG rate was higher ($P < .0001$, $P < .0001$, and $P = .03$, respectively). Similar findings were seen for structures treated to $\geq 90\%$ Dose.

Of note, in the 3-group comparison for percent treated disc or macula, the NVG risk for patients with partial or maximal sparing of the disc or macula is significantly lower than with no sparing ($P < .0001$ and $P < .03$, respectively; Table 3). We observed that NVG rate was lower in patients with partial sparing of the disc compared with patients who have no disc sparing, but this did not reach statistical significance in a pairwise comparison of only these subsets. These partially spared subsets include only up to 11% of the study cohort.

Importantly, when anterior and posterior critical structure DVH outcomes were evaluated together, NVG incidence was highest when both anterior and posterior critical structure volumes were above specified dose levels ($P < .0001$) (Fig. 3a, b). The combinations remain significant even if the paired structures are interchanged. We purposefully chose 1 anterior structure and 1

Table 2 Kaplan-Meier 5-year NVG rate by patient and tumor characteristics

Variable	n	5-y NVG rate (%)	Log-rank test
Tumor height (mm)			$P<.0001$
≤6	515	8.3	
>6-≤9	116	16.3	
>9	73	42.8	
Largest tumor diameter (mm)			$P<.0001$
≤10	312	8.0	
>10-≤16	346	14.9	
>16	46	31.8	
TNM size, AJCC 2010			$P<.0001$
T1	290	8.2	
T2	279	11.7	
T3	118	24.8	
T4	17	31.2	
Minimum distance to optic disc (mm)			$P=.002$
>3	395	8.9	
>0-≤3	169	12.5	
0	140	24.0	
Macular involvement			$P=.045$
No	492	11.1	
Yes	212	16.5	

Abbreviations: AJCC = American Joint Committee on Cancer; NVG = neovascular glaucoma.

posterior structure to graphically show that if either one can be spared, NVG risk is lowered, and if both can be spared, NVG risk is lowest.

Multivariate analysis

Using multivariate analysis, tumor height was the most important predictor of NVG (likelihood ratio test, $P<.0001$). The other significant independent factors in this order were age ($P<.0001$), %disc treated to $\geq 50\%$ Dose ($<100\%$ vs 100%) ($P=.0007$), LTD ($P=.01$), %lens treated to $\geq 90\%$ Dose (0 vs $>0\%$ - 30% vs $>30\%$) ($P=.01$), and ON length treated to $\geq 90\%$ Dose (≤ 1 mm vs >1 mm) ($P=.02$). Hazard ratios [95% CI] for NVG predictors were as follows: greater height (HR 1.09 [1.04-1.13]), older age (HR 1.01 [1.01-1.02]), 100% disc treated to $\geq 50\%$ Dose (HR 3.38 [1.63-6.97]), larger diameter (HR 1.04 [1.01-1.07]), $>30\%$ lens treated to $\geq 90\%$ Dose (HR 1.80 [1.15-2.82]), and longer length of ON treated to $\geq 90\%$ Dose (HR 2.49 [1.20-5.16]). Of note, the analysis was repeated substituting TNM size category for tumor height and LTD. Consistent conclusions regarding predictors were reached, with independent predictors of outcome in this order: % lens treated to $\geq 90\%$ Dose ($P<.0001$), TNM size category ($P=.0001$), age ($P=.0001$), %disc treated to $\geq 50\%$ Dose ($P=.0003$), and ON length treated to $\geq 90\%$ Dose ($P=.03$).

For each dose level considered in these analyses (50% and 90%) the lens and CB volume treated were highly correlated ($r=0.72$, $P<.0001$; $r=0.65$, $P<.0001$, respectively), with each factor being a significant predictor of NVG with univariate methods. With such strong correlations, it would be expected that at most only 1 of these factors would be an independent predictor of NVG. Similarly, disc dose and macula dose were highly

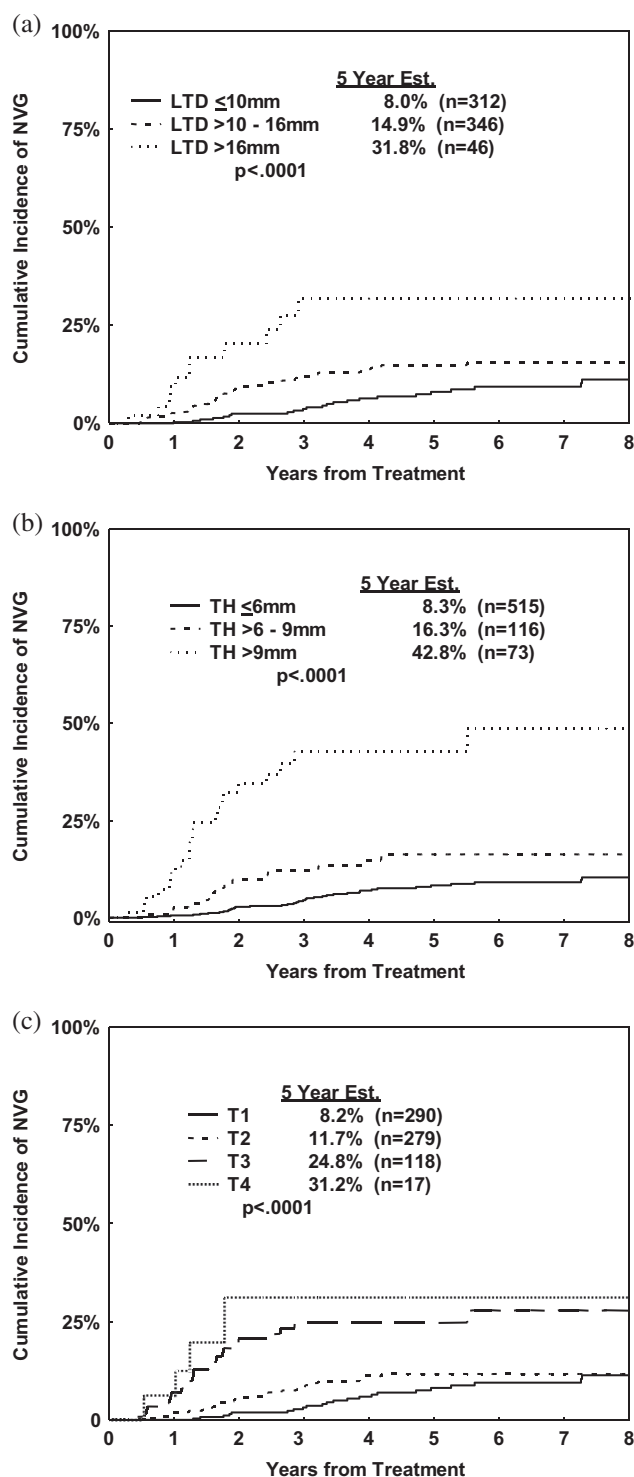


Fig. 2. Kaplan-Meier estimates of neovascular glaucoma (NVG) after proton beam therapy as a function of (a) largest tumor diameter (LTD) ≤ 10 , >10 - 16 , or >16 mm ($P<.0001$); (b) tumor height (TH) ≤ 6 , >6 - 9 , or >9 mm ($P<.0001$); and (c) TNM size category (American Joint Committee on Cancer, 2010, 7th edition) ($P<.0001$).

correlated ($P<.0001$). In addition, it was observed that patients treated to higher doses to both anterior and posterior structures (ie, lens or CB [$>30\%$] and disc [100%]) have larger tumors in terms

Table 3 Five-year Kaplan-Meier estimates of NVG for proton beam therapy patients by dose volume parameters for critical structures receiving $\geq 50\%$ Dose (equivalent of ≥ 28 GyE) and $\geq 90\%$ Dose (≥ 50 GyE)

Critical structure	NVG rate (%) by proton beam dose received			
	$\geq 50\%$ Dose		$\geq 90\%$ Dose	
	5-y estimate (95% CI)	n	5-y estimate (95% CI)	n
% Lens volume				
0	6.8 (4.6-10.1)	443	9.0 (6.7-12.2)	561
>0-30%	19.8 (14.0-27.1)	195	24.8 (16.7-35.8)	103
>30%	36.9 (24.7-54.0)	66	44.6 (25.7-69.0)	40
	$P < .0001$		$P < .0001$	
% Ciliary body volume				
$\leq 30\%$	8.7 (6.4-11.8)	547	11.6 (9.0-14.8)	627
>30%	29.4 (21.6-39.4)	157	22.4 (13.7-35.3)	77
	$P < .0001$		$P = .002$	
% Optic disc				
0	7.5 (5.2-10.2)	434	7.8 (5.5-11.0)	480
>0-<100%	12.3 (6.0-24.4)	68	19.1 (8.6-39.3)	44
100%	24.4 (18.2-32.3)	202	24.7 (18.2-32.9)	180
	$P < .0001$		$P < .0001$	
% Macula				
0	8.7 (6.0-12.7)	349	9.2 (6.4-13.0)	409
>0-<100%	14.5 (7.1-28.4)	79	14.1 (6.9-27.6)	58
100%	17.4 (12.9-23.2)	276	18.3 (13.4-24.7)	237
	$P = .03$		$P = .02$	
Length of optic nerve				
0- ≤ 1 mm	8.4 (6.0-11.7)	467	8.7 (6.3-11.9)	508
>1 mm	21.4 (15.9-28.3)	237	23.3 (17.3-31.1)	196
	$P < .0001$		$P < .0001$	

Abbreviations: CI = confidence interval; NVG = neovascular glaucoma.

P values by log-rank test.

of both height and diameter (ANOVA, $P < .001$ for each). When summarizing these results, after adjusting for tumor dimensions in a multivariate model, the NVG risk is also independently predicted by dose-volume parameters.

For this study cohort there was no difference in age by location when evaluated by disc ($P = .50$) or macular involvement ($P = .40$); but for the 72 patients (10%) with CBI, these patients were significantly older (means: 63.5 vs 59.4 years, $P = .025$). When building a predictive multivariate model, age accounted for more of the variability than CBI. Hence, with the small proportion of patients with CBI, this was not an independent contributing factor.

Discussion

Historically, studies that first analyzed the effect of charged particle therapy for ocular melanoma identified NVG as a major concern in addressing optimal treatment (7, 8). This report details the rate of NVG in our cohort of proton patients and reveals an overall low 5-year NVG rate of 12.7% compared with recent and historical results. For those who developed NVG in our cohort, the median time to NVG diagnosis was 22 months after treatment

(range, 3.6-87 months). Current literature shows overall 5-year NVG rates of approximately 15% to 31% in modern proton-treated UM patients with NVG, and this typically occurs in the first 3 years after treatment (5, 6, 12, 13). Our historical cohort of helium-ion irradiated patients revealed a higher 5-year NVG rate of 35.5% (95% CI 30.5%-41.0%), with NVG diagnosis occurring at a median of 18.4 months after treatment (range, 3.0-77.5 months), and 5-year rate of enucleation due to NVG of 12.5% (8). In our present proton cohort of patients, the 5-year rate of enucleation due to NVG was 4.9%. In the contemporary literature, resultant enucleations from complications generally at 5 years have ranged from 2.2% to 18% and are related to NVG in the majority of cases (5, 6, 12).

The pathogenesis of NVG is considered multifactorial and may include the release of angiogenic factors, posterior segment ischemia, vascular occlusion, tumor effects, radiation retinopathy, and inflammation (14). Multiple parameters may have a role in the development and/or progression of NVG, including patient, tumor, treatment, and critical structure dosing parameters. In the present study we found that greater tumor height, older age, and larger tumor diameter significantly predicted higher NVG incidence. Tumor-node-metastasis size category similarly was predictive of NVG risk. This is consistent with other particle therapy groups who have noted such tumor and patient characteristics as potentially predictive for NVG (3, 7, 8, 15). Treatment options such as local resection of tumors with radiation preceding or following may decrease NVG risk by reducing the tumoricidal load and inflammatory factors (16). The management of NVG has also evolved and markedly improved during the course of the study time period. Anti-vascular endothelial growth factor therapy and photocoagulation after particle therapy have a significant role in the treatment of NVG and likely reduce the rate of enucleation due to NVG (17). The efficacy and optimum use of these agents has not been completely delineated.

Along with tumor, patient, and other surgical and therapeutic factors, specific RT parameters like total dose, fractionation, and biologic effectiveness may be significant in the development of NVG. In fact, there was a 3-fold decrease in NVG incidence with a decrease in dose from 80 GyE to 60 GyE in helium-treated patients (18). Because dose was optimized at 56 GyE in 4 fractions historically at our institution (unpublished results), other parameters were considered for adjustment to potentially further reduce NVG side effects while maintaining high tumor control; this included techniques for sparing critical structures.

Decrease in NVG with anterior structure-sparing techniques

Prior studies have shown that dose to the anterior segment may be a risk factor for NVG (3, 6-8). Because an active attempt is made to spare the anterior segment by adjusting the eye fixation angle primarily, only 9.2% of our PBRT patients had >30% anterior structure (CB and lens) receiving $\geq 50\%$ Dose. Among our patients with higher doses to the CB and/or lens, the NVG rate was significantly increased ($P < .0001$). In terms of margin, Egger et al (19) found that a reduced aperture safety margin below 2 mm resulted in decreased local control and abandoned reduced safety margins <2 mm. We use a very focused aperture safety margin reduction from 2.5 mm to 2 mm when appropriate to limit dose to a critical structure and similarly do not decrease aperture margin below 2 mm in any case.

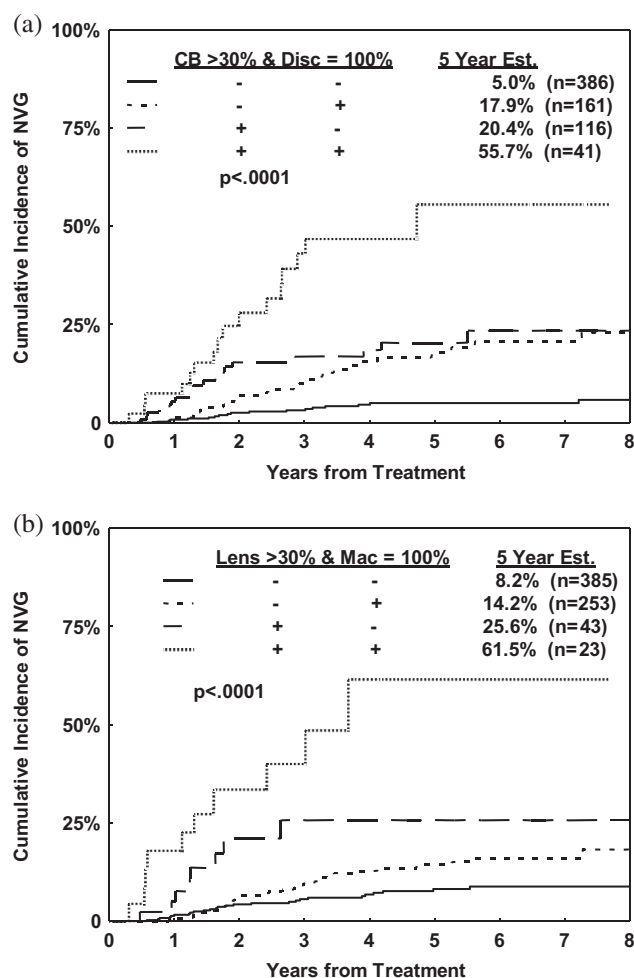


Fig. 3. Kaplan-Meier neovascular glaucoma (NVG) estimates after proton beam therapy as a function of anterior and posterior structure dose volume parameters with (a) %volume of ciliary body (CB) and %disc receiving $\geq 50\%$ Dose ($P < .0001$); and (b) % volume of lens and macula receiving $\geq 50\%$ Dose ($P < .0001$). In (a) and (b), anterior dose volume cut point “+” signifies that $>30\%$ of the CB or lens volume received $\geq 50\%$ Dose (28 GyE or higher). Posterior dose volume cut point “+” signifies that 100% of the optic disc or macula received $\geq 50\%$ Dose. When both factors were at increased levels, NVG incidence was highest. When either anterior or posterior dose only was at or above the specified cut points, NVG occurrence was intermediate. When neither dose volume cut point was met, then NVG incidence was lowest.

Dendale et al (12) found that rate of enucleation due to complications was related to tumor thickness and volume of lens treated ($\geq 43\%$ receiving ≥ 30 GyE). In the present study, the data show a relationship between the extent of normal tissue irradiation and the extent of NVG. Dose–volume histogram analysis confirms that when $>30\%$ of the anterior critical structures (lens or CB) received $\geq 50\%$ Dose (≥ 28 GyE), there was a substantially higher probability of NVG ($P < .0001$). These parameters may assist in clinical practice to help in evaluating NVG risk and as potential dose guidelines for planning purpose. The challenge remains to determine the exact cause (ie, direct injury, tumor effects, inflammation, or other causes); and although a relationship

is observed in this study cohort between critical structure dose and NVG risk, causation cannot be proven.

Posterior structure dose and NVG

The distance to the disc and resultant dose have been identified as potential risk factors for NVG (3, 15). Patients treated in Japan for unfavorably located and larger UM tumors by carbon ion with and without optic disc irradiation have been shown to have 3-year NVG rates of 62.9% versus 28.4%, respectively (20). In the present study, multivariate analysis indicated that significant predictors of NVG included the percent treated disc and nerve. The dose to these structures is clearly related to the tumor location. Major branches of the central retinal artery may be occluded by both tumor location and radiation to this area, thus potentially leading to increased posterior segment ischemia. Unlike minimizing the dose to the CB and lens by changing beam entry angle, when treating posterior segment tumors it is often not possible to decrease dose to the disc or macula owing to the presence of tumor in this region, although a portion of the structure may be spared by careful beam shaping.

Conclusion

Charged particle therapy has been a mainstay of treatment for UM, although NVG has historically been a major complication. This study provides evidence that the overall occurrence of NVG and enucleation due to NVG are low compared with historical reports. The data support that tumor size, as well as critical structure dosing, are both independently predictive of NVG. Dose–volume histogram analysis showed that when both anterior and posterior critical structure doses reached certain dose levels the NVG incidence was highest, compared with if anterior or posterior or both dose levels could be lowered. The relationship between NVG and critical structure doses has supported the role for tissue sparing in treatment planning when appropriate and possible, to potentially lower the overall risk of NVG-related morbidity. Although individual variability exists, clinically the specified tumor and dose–volume parameters potentially help in further evaluating NVG risk.

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